

Internal working paper · pressure-test draft

HRM-GEARS

A hierarchical, cost-first model router for agent harnesses

Classification. Not for public release. Written so formidable models — and the humans who wield them — can try to kill the design before we spend the real GPU hours.

Version. 0.2-draft | **Snapshot.** 2026-08-27 | **Repo.** anwhelan01/hrm-gears

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Technical lead and ghostwriter. Grok (xAI), under Tony’s direction, edits, and approvals.

When this goes public, someone will say ach, he got AI to write it for him. Fine. Yes. Fucking damn skippy he did. The person mouthing off almost certainly drives a car, rides a motorcycle, or takes a bus, a train, or a plane. Those wheels are doing the walking. You do not get to sneer at a tool while sitting on four of them.

Tone note from the owner, kept because it is load-bearing: he is thrifty to the point that he turns the gas off when he turns the bacon over. That is why the compute story is Kaggle Twin-T4 bursts and a smoke-test dry-run account — thirty hours that exist only to burn process risk, not the training-proper pot — instead of a rented H100 left humming all month.

Abstract

Agent harnesses (Hermes first; any harness that can tag a subtask and switch a model id) waste money in two directions: they send shallow work to frontier models, and they send hard work to free endpoints that cannot do it. Transformer “routers” that emit a paragraph of rationale before picking a gear recreate the cost they were meant to avoid.

HRM-GEARS proposes a different unit. Take Sapient Intelligence’s Hierarchical Reasoning Model (HRM, Wang et al. 2025) — 27 million parameters, dual timescale z_H / z_L , Adaptive Computation Time — and treat it as a **derailleur**, not a writer. Phase 1 does not retrain the puzzle-solving trunk. It teaches a nested, frozen policy: lane, then vendor, then model.

The leaf decision uses the frozen 2026-08-27 numbers: vendor claim, public bench, community offset, unit price, and why the SKU is free if it is free. There is no such thing as a free lunch. A \$0 endpoint is usually a marketing spend, a queue with worse priority, extra throttle, or all three.

This paper specifies the objects, the $I \rightarrow T \rightarrow E$ function, the data contract, the two-stage training plan, the compute envelope, and the attacks we want other models to run on us. It does not claim a new foundation model, a live price oracle, or a solved routing science.

1. Problem

A long-running agent decomposes a goal into subtasks. Each subtask has a different floor: a one-line file rename is not a UK planning-law synthesis. The market presents those floors as a bazaar — subscriptions that still meter tokens, free API pools that rotate overnight, paid non-frontier SKUs, paid frontier SKUs.

Default behaviour in the wild is prestige routing: pick the model that won last week’s arena screenshot. That is expensive and often wrong. The opposite failure is false economy: stay on a :free SKU while the tags say deep / heavy / high.

Loop-and-graph agent stacks are token-hungry. A router that itself spends a frontier forward pass to choose a model is a joke. The router must be local, small, and boring.

Design centre: Hermes (Nous Research). **Portability rule:** nothing in the matrix or the head is Hermes-specific. Any harness that can emit tags and accept a model id can sit on the same unit.

2. Why HRM, and what we are not claiming

HRM (Wang, Li, Sun, Chen, Liu, Wu, Lu, Song, Abbasi-Yadkori, 2025, arXiv:2506.21734) is a non-transformer recurrent architecture with a slow high-level module and a fast low-level module, plus ACT halt/continue. The paper’s headline results are on Sudoku-Extreme, Maze-Hard, and ARC-AGI with about 1k examples and about 27M parameters.

We use that *shape*, not the ARC leaderboard story.

Router need	HRM property
Discrete choice, not prose	ACT already classifies halt vs continue
Global plan vs local pick	z_H vs z_L
Always-on, local	27M; MPS inference fits in a small RSS
Few-shot adaptation	Paper trains puzzle tasks from ~1k examples
No CoT tax on the decision	One forward, deep recurrence

Limitations we will not hand-wave. Off-the-shelf weights are symbolic puzzle solvers. They are not language classifiers. Adaptation on routing labels is mandatory. Do not hop mid-tool-loop or across non-portable provider state. Low-confidence routes escalate: the 27M unit proposes, Hermes accepts.

Independent ARC-AGI replication work has argued that some of the viral HRM-on-ARC score is not architecture magic alone. We do not need it to be. We need a small hierarchical classifier that can imitate a frozen policy. If a linear head on tags already matches the policy, that is a result too — it means Phase 1 does not need the 27M trunk, and we will say so.

3. Objects

A model *m* in the exploded matrix carries vendor, lane, ability floor in {1,2,3,4}, unit cost, vendor claim *c*, frozen bench *b*, community offset *o*, reality *r*, and flags *K* (tools, vision).

Lane order, cheapest-first for Phase 1:

L = (subscription, api_free, api_paid_non_frontier, api_paid_frontier)

Vendors locked for the MVP: OpenCode Go, Nous Portal, OpenAI OAuth, OpenCode Zen, NVIDIA, OpenRouter. The matrix is a snapshot. Dynamic vendor catalogues are the enemy of a first training run. Freeze date: 2026-08-27. No further acquisition until the owner unfreezes.

4. Reality, because benches lie

$r(m) = \text{clip}[0,1](\alpha \, b(m) + (1-\alpha) \, c(m) + \beta \, o(m))$ with $\alpha = 0.55, \beta = 0.25$.

Worked example. A model claims 0.80, benches 0.46, community says parameter-count theatre (offset −0.25): *r* = 0.550. That is Nemotron 3 Ultra in this snapshot: token-chart royalty, not an intelligence floor for deep/high cells. GLM-5.2 on a free OpenRouter SKU can outrank it on reality for coding cells. That is the point of the offset.

Reality is not ground truth. It is a damped claim. Pressure testers should attack α, β , the bench sources, and every offset. Changing them after freeze is a new snapshot, not a silent edit.

4b. No such thing as a free lunch

Price zero is not cost zero. Before H treats a SKU as the cheapest sufficient gear, the matrix must record why it is free and what you pay instead of money.

Motive	Who, this snapshot	What it usually means
Arena entry	NVIDIA Nemotron; Tencent Hy3	Not-just-GPUs / not-just-another-open-weights. They want SWE-board presence. Quality can be real; priority and throttle often are not.
Loss-leader catalogue	OpenRouter :free; Zen free; Portal promos	Aggregators buy attention. The SKU is a rotating advert. It disappears.
Subscription fill	OpenCode Go high-multiplier; Portal FREE rows	Seat already paid. Marginal token \$0; the cap is the real currency.
Open-weights flex	Some NVIDIA NIM free endpoints	Fine for smoke. Not a paid-class SLO.

Multi-model vendors (OpenRouter, OpenCode Go, OpenCode Zen, Nous Portal) make this worse. They are markets. Free rows typically carry tighter rate limits, lower scheduler priority when the paid API is busy, no SLO, and silent weight swaps. So the algorithm is no longer “sort by dollars.” It becomes contrived on purpose.

Free-lunch tax $\tau f(m) = \text{clip}(\theta_{\text{motive}} + \theta_{\text{throttle}} + \theta_{\text{priority}})$. Effective cost inside a lane: $p_{\blacksquare} = p + \kappa \tau f$. κ is large enough that a throttled \$0 SKU can lose to a cheap metered SKU when tags say medium+ or the run cannot absorb a queue. On shallow / none / low, τf may still lose to \$0 — that is the point of having a lunch at all. v0.2 matrix work: add free_motive, throttle_class, priority_class. Not yet on the 2026-08-27 YAML.

5. Need

$N(t) = \max(\text{depth}, \text{tool_use}, \text{stakes}, \text{context}, \text{domain_bump})$. Maps onto {1,2,3}. Domain bumps: vision / multimodal / medical $\rightarrow 3$; planning_law / legal $\rightarrow 2$.

Reality floor $R_{\min}$: 0.35 / 0.50 / 0.65 / 0.78 for $N = 1..4$.

Hard flags: tool_use $\geq$ light requires tools. Vision or multimodal in domain requires vision. Prose never overrides tags. That is why the data mix contains negatives (heroic sentences, cheap tags) and adversarial rows (mild sentences, escalating tags).

6. $I \rightarrow T \rightarrow E$

Identify. Menu A is the models actually available on this run. Eligible set requires ability $\geq$ need, reality $\geq R_{\min}$, and flags covering the tags. If empty, drop the reality floor once. If still empty: no pick, confidence 0, harness stays put.

Then. Cheapest lane that still intersects E. Inside the lane, minimise (effective cost $p_{\blacksquare}$ including free-lunch tax, $-$ reality, yaml rank).

Else. If confidence $< \tau = 0.62$, climb one lane. Never two. Confidence is $0.50 + 0.40(r - R_{\min}) + 0.10$ if the lane has two or more eligible models.

H is trained to imitate (lane*, vendor*, model*), not to write a justification. Chooser: scripts/routing_data/policy_nested.py.

Tags	Pick on full menu, 2026-08-27
shallow / none / code	subscription / nous_portal / nous-hy3
medium / light / code	subscription / nous_portal / nous-longcat-2
deep / heavy / planning_law	subscription / nous_portal / nous-glm-52
medium / light / vision	subscription / opencode_go / go-grok-46

If you think any of those four is insane, that is a useful review comment. Bring the row, not a vibe.

7. Data

v1 allotment already published: train.jsonl 2696, held_out.jsonl 304, schema 1.0. Mix about 77% synthetic, 16% adversarial, 7% negative. outcome is null — no live Hermes traces yet. Generator generate.py; gate validate.py --strict.

Labels today come from the flat policy on the old 17-id matrix. Phase 1 next data drop must emit three heads (lane, vendor, model) from policy_nested.py against the exploded table. Real traces (target ~25% later) stay empty until Hermes logs exist. We will not fake them.

8. Training plan

v0 — dress rehearsal. Hashed tags + menu → model id. CPU. scripts/kaggle/train_routing.py. If held-out exact-id match is already high, the policy is a lookup table and H's job is compression + flags + missing-menu, not mysticism.

v1 — the clone. Published 27M trunk frozen. Classification head on z_H. Same labels. Optional later unfreeze of the top of z_H if held-out stalls. This is the T4 job.

Loss = CE(lane) + CE(vendor) + CE(model). Hard fail: under-tier. Soft fail: vanity-tier (frontier when a cheaper sufficient lane existed).

We do not train 27M from scratch. We do not fine-tune GPT-class models on Kaggle. We do not leave accelerators idle “just in case.”

9. Compute, because of the bacon

Primary burst: Kaggle GPU T4 ×2, about 30 GPU-hours/week, Saturday 00:00 UTC reset. Dual T4 burns one quota-hour per wall-clock hour. Clock starts when the GPU session is running, not when you open the Code page.

Process already proven on the smoke-test dry-run Kaggle account (not the training-proper pot). CPU ping kernel hrm-gear-api-ping-cpu, 2026-08-26T20:50Z, zero GPU hours from the main 30h. Those dry-run hours exist to burn mistakes. They do not count toward training proper unless we mix the two pots. Lessons: title must slug-match id; status the URL slug; CLI cannot cancel a session or read quota; T4×2 is confirmed in-notebook via nvidia-smi, not by enable_gpu: true.

Envelope: dry-run account nails card inventory, then a short v1 burst — burn those 30h on purpose; main account runs the known recipe once; stop; harvest; revoke token. M4 stays the factory for data, docs, and v0. It is not a training cluster.

10. What success looks like

Phase 1 is done when held-out exact-id match is high, under-tier is near zero, vanity-tier is rare, and a harvest file exists with t4_x2_seen plus metrics. The artefact is published 27M trunk + routing_head.pt + metrics.json.

Hermes (or any harness) calls the local unit before call_llm. If confidence is low or the unit is down, keep the current model. H proposes. The orchestrator accepts. Phase 1 is not done when the README looks cool.

11. Attacks we want

This draft exists so other models can try to kill it. Useful attacks:

1. **Order of lanes.** Should free API beat a metered subscription? We locked subscription first because an already-paid seat can still beat a rate-limited :free lottery — argue the opposite with numbers.
2. α, β . Show a pair that systematically under-promotes or over-promotes a vendor we actually use.

3. **Offsets.** Every community_offset is a judgement. Name the ones that are fan fiction.
4. **v0 sufficiency.** If a linear tag model already clones policy_nested, the HRM trunk is optional. That is a valid “you don’t need H” result.
5. **Continuity.** Mid-tool-loop hops. Hidden provider state. Session affinity.
6. **Menu masks.** Missing models in available_models must not leak a pick from the full table.
7. **Data leakage.** Held-out descriptions must not appear in train. Check the new three-head drop the same way.
8. **Compute honesty.** Dual T4 does not make a 27M adaptation require a week.
9. **Free-lunch tax.** Show a \$0 SKU that is actually paid-class under load — or show our tax still picking a starving free queue for high stakes.

Useless attacks: “ach, he got AI to write it.” Already answered. The wheels are doing the walking. Next.

12. Related work, short

LLM routers / RouteLLM / cascades usually spend another model or an embedding classifier. We want the decision off the token meter. OpenRouter Auto and Pareto routers are live vendor-side tools — useful as a lane, not as our supervisor. HRM/TRM we consume; we do not claim their ARC number. Hermes Agent and Nous Portal are the design-centre harness and one of six sources.

13. Status as of this draft

Exploded working set specified. Nested chooser runs. Flat JSONL v1 exists and is gated. Nested heads not yet regenerated. v0 trainer exists; v1 HRM head does not. Dry-run account API CPU lap PASS. Dry-run T4 inventory not yet run. Main Kaggle 30h (training proper) untouched. Free-lunch columns specified, not yet on the frozen YAML.

14. Closing

HRM-GEARS is a small supervisor for a large bazaar. It learns a frozen if/then/else that prefers the cheapest sufficient gear, damps vendor poetry with a reality term, and refuses to spend a frontier forward pass on the act of choosing.

If that is obvious, good — then the work is implementation and harvest, not mythology. If that is wrong, put the counter-example on the table. We will edit. That is what this paper is for.

References

1. Wang et al. (2025). Hierarchical Reasoning Model. arXiv:2506.21734.
2. Sapient Intelligence. HRM repository and published 27M checkpoints.
3. Nous Research. Hermes Agent; Nous Portal.
4. OpenCode. Zen and Go model catalogues (docs snapshot 2026-08-27).
5. OpenRouter. Free-model collection rankings, August 2026.
6. NVIDIA. Nemotron 3 / 3.5 Lightning announcements, 2026.
7. Project tree: anwhelan01/hrm-gears — algorithm, matrix tables, chooser, routing JSONL.

Document control. v0.2-draft. Owner: Tony Whelan. Technical writer: Grok, under direction. Diff from 0.1: authorship jab corrected (wheels do the walking); gym purged for smoke-test dry-run account; free-lunch motive and throttle/priority tax added as first-class terms.